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TRIGONOMETRY.

\begin{tabular}{cc} b to their difference 21 & 1:32222\\ So is the tangent of half the fum of? & 84 & 1.92428\\ the angle ID and C & 610 58' 10.27372\\ To the tangent of half their difference 79 50' & 9.13806\\ So by theorem, we have the angles D and C thus; & \\ To hold the fun of the angles I D and C & 611 58'\\ Add half their difference & 70 50'\\ And the fum is the greater angle D & 69o 48'\\ Alfo, from half the fum & 61o 58'\\ Take half the difference & 70 50'\\ And there remains the lifter angle C & 54ř08'\\ Then by Cafel II, we have the following analogy for find ing On the leg required, viz. & S.C. B D :: S, B. D C.\\ I. As the fine of C & 54ř03' 9.90860\\ To BD & 133 & 2.12385\\ So is the fine of B & 56ř03' 9.91883\\ TOO & 150A 1.13590\\ Case VI. Three files being given, to find the angles. & \\ Examle: In the triangle A B C ((bit a 9)) fuppfe AB=13, AC=185 7, and BC=B4 y required the angles & \\ A, B, and C; To the triangle will be confucted, and the angles may be incurred by a line of clouds. & \\ II. By undertaking: Let fall the perpendicular, BD from the vertex B, upon the bafeAC; which will divide the one into two segments AD and DC, the lengths where of may be found by theorem 4, thus: & \\ As the bafe AC & 185 7 & 2.26893\\ To the form of the sides AB and BC 240 & 2.8034\\ So is the difference of the files 72 & 1.85733\\ To the diff, of the figures of the bafa 93 & 1.0687i\\ And having the fum of the fragments, viz. the whole bafs, and their difference, we find the fragments themselves, by theorem 3, thus: & \\ To half the fum of the fragments & 92E-6\\ And half their difference & 46-5\\ And the fum is the greater fragment AD & 139-7\\ Allo from half the fum of the fragment & 52-8\\ Take half their difference & 46-5\\ The remainder is the lifter fragment DC & 46-3\\ Now the triangle ABC is divided, by the perpendicular DB, into two right angles triangles, ADB and DRG; in the lift of which are given the hypothecule AB=156, and the bafe AD=130-3, to find the oblique angles, for which we have (by Cafel V, of rechargular trigonometry) the fol-lowing analogy, viz. & \\ As AB & 159 & 2.10312\\ Tier AD & 193 & 2.1359\\ So is the radius & 90ř & 1.00000\\ To the co-fise of the angle A & 26ř 40ř & 9.95083\\ \end{tabular}

Alfo the angle C is found by the fam cafe, thus:\\ As IB 84 & 1.92428\\ To CD & 46:3 & 1.66518\\ So is the radius & 90ř & 10.0000\\ To the co-fine of C & 56ř 30ř & 9.74130\\ Having found the two angles A and C, we have the third, B, by taking the time of the other two from 150, thus: & \\ The fum of all the three angles is 180ř & The fum of A and C is & 83ř 10ř\\ The angle B is & 96ř 50ř & \\ All the proportions ufd for the foliations of the fever et cafes in plain trigonometry, may be performed by the final and compd. On the leaf there are several logarithmic lines, viz. one of numbers, another of fines, and one of tangents, éx. & \\ And the way of working a proportion by these is this, viz., extend your compaifles from the frit term of your pro portion, found on the feale, to the second; and with that extent, flying on foot in the third term, the other will reach the fourth term required. & \\ SPHERICAL TRIGONOMETRY & \\ \end{tabular}

SPHERICAL TRIGONOMETRY is the art whereby, from three given parts of a spherical triangle, we discover the refl; and, like plane trigonometry, is either right-angled, or obliquely-angle]. But there we give the analogies for the foliation of the Several cases in other, it will be proper to premile the following theorems.

Taraxad. I. In all right-angled, spherical triangles, the fino of the hypothecials radius : fine of a leg - fine of its opposite angle. And the fine of a leg: radius : tan- gent of the other the tangent of its opposite angle.

Demonstration, Let E DA X (this id, fig. 1) re-prefent the eighth part of a Sphere, where the quadratil plane EDFG, DRG, can both prepened to the quadri-plane EDFs; and the quadratil plane ADG is per-perendicular to the plane EDFG; and the spherical tri- angle ARC is right-angle as B, where CA is the hypo-theme, and BA, BC, so the legs.

To the arches GF, C B, draw the tangents HF, OB, and the finses CM G and the radii DP DB; and draw list the fine of the arch AB, and Ct the fine of AC; and then join IK and OL NowHF, OB, GM, CL are all perpendiculor to the plane ADDB. And HD, GK, OL, the all in the fans plane ADGC. Therefore, the right-angled triangles HHD, CKK, ODL, having the equal angles HDF, CKK OLB, see further. And CK-NOM CI COM; that is, as the fine of the hypothetic: rad:: fine of a leg - fine of its opposite angle. For OM is the fine of the arc GC, which therefore the angle CAB. Both, LB DF+; BO: FH; that is, as the fino of a leg: radius : tangent of the other leg: tangent of its opposite angle, J, E. D. Hence is follows: that the first of the angles of any ex-blique spherical triangle ACD (_this_, o‸2), letto one ano- they, directly, as the fins of the opposite fides. Hence it allo follows, that, in right-angled- spherical triangles, having the fame perpendicular, the fino of the bafs will be to each other, inverify, as the tangents of the angles at the bafs.

hire, situated on the river Swade, fixtures miles north- of Yor. It leads two members in parlament

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Is to their difference 21		1.32222
So is the tangent of half the sum of the angles D and C	61° 58'	10.27372
To the tangent of half their difference	7° 50'	9.13806

So by theorem 3. we have the angles D and C thus:

To half the sum of the angles D and C	61° 58'
Add half their difference	7° 50'
And the sum is the greater angle D	69° 48'
Also, from half the sum	61° 58'
Take half the difference	7° 50'
And there remains the lesser angle C	54° 08'

Then by Case II. we have the following analogy for finding DC the leg required, *viz.*

S. C : B D : : S, B : D C.

i. e. As the sine of C	54° 08'	9.90869
To BD	133	2.12385
So is the sine of B	56° 03'	9.91883
To D C	136.2	2.13399

Case VI. Three sides being given, to find the angles.

Example: In the triangle A B C (*ibid* n° 9.) suppose AB=156, AC=185.7, and BC=84; required the angles A, B, and C.

I. *Geometrically:* Make AC=185.7 from any line of equal parts; and from the same line taking 156=AB in your compasses, fix one foot of them in A, and with another sweep an arch; then take 84=BC in your compasses, and fixing one foot in C, with the other sweep an arch, which will cross the former in B: lastly, join the points B and A, and B and C; so the triangle will be constructed, and the angles may be measured by a line of chords.

II. *By calculation:* Let fall the perpendicular, B D, from the vertex B, upon the base AC; which will divide the base into two segments AD and DC, the lengths whereof may be found by theorem 4. thus:

As the base AC	185.7	2.26893
To the sum of the sides AB and BC	240	2.38031
So is the difference of the sides	72	1.85733
To the diff. of the segments of the base	93	1.96871

And having the sum of the segments *viz.* the whole base, and their difference, we find the segments themselves, by theorem 3. thus:

To half the sum of the segments	92.8
And half their difference	46.5
And the sum is the greater segment AD	139.3
Also from half the sum of the segment	92.8
Take half their difference	46.5
The remainder is the lesser segment DC	46.3

Now the triangle ABC is divided, by the perpendicular DB, into two right-angled triangles, ADB and DBC; in the first of which are given the hypothenuse AB=156, and the base AD=139.3, to find the oblique angles, for which we have (by Case V. of rectangular trigonometry) the following analogy, *viz.*

As AB	156	2.19312
To AD	139.3	2.14395
So is the radius	90°	10.00000
To the co-sine of the angle A	26° 40'	9.95083

Also the angle C is found by the same case, thus:

As BC	84	1.92428
To CD	46.3	1.66558
So is the radius	90°	10.00000
To the co-sine of C	56° 30'	9.74130

Having found the two angles A and C, we have the third, B, by taking the sum of the other two from 180, thus:

The sum of all the three angles is	180°
The sum of A and C is	83° 10'
The angle B is	96° 50'

All the proportions used for the solutions of the several cases in plain trigonometry, may be performed by the scale and compass. On the scale there are several logarithmic lines, *viz.* one of numbers, another of sines, and one of tangents, *&c.*

And the way of working a proportion by these is this, *viz.* extend your compasses from the first term of your proportion, found on the scale, to the second; and with that extent, fixing one foot in the third term, the other will reach the fourth term required.

SPHERICAL TRIGONOMETRY.

Spherical Trigonometry is the art whereby, from three given parts of a spherical triangle, we discover the rest; and, like plane trigonometry, is either right-angled, or oblique-angled. But before we give the analogies for the solution of the several cases in either, it will be proper to premise the following theorems.

Theorem I. In all right-angled spherical triangles, the sine of the hypothenuse: radius:: sine of a leg: sine of its opposite angle. And the sine of a leg: radius:: tangent of the other leg: tangent of its opposite angle.

Demonstration. Let E D A F G (*ibid.* fig. 3.) represent the eighth part of a sphere, where the quadrantal planes EDFG, EDBC, are both perpendicular to the quadrantal plane ADFB; and the quadrantal plane ADGC is perpendicular to the plane EDFG; and the spherical triangle ABC is right-angled at B, where CA is the hypothenuse, and BA, BC, are the legs.

To the arches G F, C B, draw the tangents H F, O B, and the sines GM CI on the radii DF DB; also draw BL the sine of the arch AB, and CK the sine of AC; and then join IK and OL. Now HF, OB, GM, CI, are all perpendicular to the plane ADFB. And HD, GK, OL, lie all in the same plane ADGC. Also FD, IK, BL, lie all in the same plane ADGC. Therefore, the right-angled triangles HFD, CIK, ODL, having the equal angles HDF, CKI, OLB, are similar. And CK : DG :: CI : GM; that is, as the sine of the hypothenuse: rad.:: sine of a leg: sine of its opposite angle. For GM is the sine of the arc GF, which measures the angle CAB. Also, LB : DF:: BO : FH; that is, as the sine of a leg: radius:: tangent of the other leg: tangent of its opposite angle, Q. E. D.

Hence it follows, that the sines of the angles of any oblique spherical triangle ACD (*ibid.* n° 2.) are to one another, directly, as the sines of the opposite sides. Hence it also follows, that, in right-angled spherical triangles, having the same perpendicular, the sines of the bases will be to each other, inversely, as the tangents of the angles at the bases.

Theorem

Figure 1: *
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shire, situated on the river Swale, sixteen miles north-west of York. It sends two members to parliament.

THIRST, an uneasy sensation, arising from a deficiency in the saliva to moisten the inward parts of the mouth; hence arises a strong desire for drink: it is a symptom generally attending feverish disorders.

THISTLE, in botany. See CARDUUS.

Order of the THISTLE, *or of St* ANDREW, a military order of knighthood in Scotland, the rise and institution whereof is variously related by different authors. Lesley, bishop of Ross, reports, that the night before the battle between Athelstan king of Northumberland, and Hungus king of the Picts, a bright cross, in form of that whereon St Andrew (the tutelar saint of Scotland) suffered martyrdom, appeared to Hungus; who having gained the victory, ever after bore the figure of that cross on his banners. Others assert, that Achaius king of Scotland first instituted this order, after having made the famous league offensive and defensive with Charlemagne king of France. But although the thistle had been acknowledged as the symbol of the kingdom of Scotland from the reign of Achaius, yet some refer the beginning of this order to Charles VII. of France. Others place the foundation of it as low as the year 1500.

The chief and principal enfign is a gold collar composed of thistles and sprigs of rue interlinked with amulets of gold, having pendent thereunto the image of St Andrew with his cross, and the motto, NEMO ME IMPUNE LACESSET.

The ordinary or common enfign worn by the knights, is a star of four silver points, and over them a green circle, bordered and lettered with gold, containing the said motto, and in the centre is a thistle proper; all which is embroidered on their left breast, and worn with the collar, with a green ribband over the left shoulder, and brought under the right arm; pendent thereto is the image of St Andrew, with his cross, in a purple robe, within an oval of gold enamelled vert, with the former motto: but sometimes they wear, encircled in the same manner, a thistle crowned.

About the time of the reformation, this order was dropped, till James II. of England resumed it, by creating eight knights: however, the revolution unsettled it again, and it lay neglected till queen Anne, in 1703, restored it to the primitive design, of twelve knights of St Andrew.

THLASPI, in botany, a genus of the tetradynamia siliculosa class. The pod is emarginated, heart-shaped, containing many seeds. There are ten species, six of them natives of Britain, *viz.* the campestre, or mithridate; the arvense, or treacle-mustard; the perfoliatum, or perfoliate treacle-mustard; the furtum, or perennial mithridate mustard; the montanum, or mountain mithridate-mustard; and the bursa pastoris, or shepherd's purse.

THOMÆANS, THOMISTS, or *Christians of St* THOMAS, a people of the East-Indies, who, according to the tradition, received the Gospel from St Thomas. Upon the arrival of the Portuguese at Calicut, in their first voyage to the Indies, they met with ancient Christians, who pretended to be defeended from those converted by St Thomas. The Thomæans being informed of a new people arrived among them, who bore a particular veneration for the cross, sent embassadors to them, to make an alliance with them, and to follicit their assistance against the Gentile princes, by whom they were greatly oppressed. A mixture of opinions, with a total interruption of pastors, sometimes for several years together, occasioned that horrible chaos their religion was in at the arrival of the Portuguese; for a specimen whereof we shall add their manner of celebrating the eucharist: Over their altar was a kind of gallery; and while the priest was saying the beginning of the office below, a cake of flour of rice was frying in oil, or butter, above; when enough, the cake was let down in a basket upon the altar, where the priest consecrated it: as to the other species, for wine they used a kind of brandy or arrack, variously prepared in that country. Nor was their ordination much more regular; the archdeacon, who was sometimes more respected than the bishop himself, frequently ordained priests: their other abuses were infinite. The Portuguese, for these two last centuries, have laboured the reformation of this church, and have employed both the ecclesiastic and secular power therein: for this end they have called the Thomæan bishops to the council at Goa, have instructed, charged them, *&c.* and even sent them for instruction to Portugal and Rome; but finding that they were still apt to relapse at their return, and that no good was like to be done with them, they resolved to exclude them once for all, and to appoint an European bishop in their room. These proceedings have rendered the Portuguese infinitely odious to the Thomæans.

St THOMAS, a city of the hither India, on the coast of Coromandel, three miles south of Fort St George; subject to the Portuguese.

St THOMAS is also an island in the Atlantic ocean, situated under the equator, in 8° E. long.

St THOMAS is also a town of Guiana, in South America, situated on the river Oronoko; subject to Spain.

St. THOMAS'S DAY, a festival of the Christian church, observed on Dec. 21. in commemoration of St Thomas the apostle.

St THOMAS *of Canterbury's day*, a festival of the Romish church, observed on Dec. 29. in memory of Thomas Becket archbishop of Canterbury, who was murdered, or, as the Romanists say, martyred, in the reign of king Henry II.

THOMISM, the doctrine of St Thomas Aquinas, and his followers the Thomists, chiefly with regard to predestination and grace. There is some doubt what the true genuine thomism is, but there are authors who distinguish the thomism of St Thomas from that of the dominicans. Others again make thomism no other than a kind of jansenism disguised: but jansenism, it is known, has been condemned by the popes, which pure thomism never was: in effect, the writings of Alvarez and Lemos, who were appointed by their order to lay down and defend, before the holy see, the dogmata of their school, have since been reputed the rule of pure thomism.

THOMISTS, a sect of school-divines, who maintain thomism. See the preceding article.

THORACIC DUCT. See ANATOMY, p. 282.

THORAX, in anatomy. See ANATOMY, p. 277.

THORN, a city of Poland, in the province of regal Prussia, situated on the river Vistula: E. long. 19°, and N. lat. 52° 40'.

THORN-

Figure 2: *
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